

Literature Benchmark Brief: Raissi_2019_JCP

Title: Physics-informed neural networks: A deep learning framework for solving forward and inverse problems

Authors: M. Raissi, P. Perdikaris, G. E. Karniadakis (2019)

Venue: Journal of Computational Physics | **DOI:** 10.1016/j.jcp.2018.10.045

Cluster: Cluster E: Missing Data & Modular Risk Layers

Core Theoretical Mechanism & Model Formulation

Automatic differentiation to penalize PDE residuals in deep neural networks

Empirical Scale & Experimental Dataset

Navier-Stokes, Burgers, and Schrödinger equations

Relevance to IEEE TSTE & Sustainable Energy Systems

Foundational paper of PINNs; informs soft penalty formulations for physical consistency

Identified Gaps & Operational Vulnerabilities

PDE residuals assume continuous differentiable domains; invalid at discrete control cliffs

Role in Manuscript Defense & Reviewer #2 Proof

Explains why aerodynamic control cliffs require state-conditional anchors rather than PINN PDEs